Chapter 8

Ratio and Proportion

8.1 Ratio

When two or more similar quantities are compared, then to represent this comparison, ratios are used.

or

Ratio of two quantities is the number of times one quantity contains another quantity of same kind.
The ratio between x and y can be represented as $x : y$, where x is called **antecedent** and y is called **consequent**.

$$\frac{x}{y} \text{ or } x : y$$

For example: There can be a ratio between ₹100 and ₹500, but there cannot be the ratio between ₹100 and 500 apples. Hence, the units of quantity for the comparison of ratio should be same.

8.1.1 Types of Ratio

The different types of ratio are explained as under

1. **Duplicate Ratio:** If two numbers are in ratio, then the ratio of their squares is called duplicate ratio. If x and y are two numbers, then the duplicate ratio of x and y would be $xr : yr$.
   For example: Duplicate ratio of $3 : 4 = 3 \times 3 : 4 \times 4 = 9 : 16$

2. **Sub-duplicate:** Ratio If two numbers are in ratio, then the ratio of their square roots is called sub-duplicate ratio. If x and y are two numbers, then the sub-duplicate ratio of x and y would be $\sqrt{x} : \sqrt{y}$.

3. **Triplicate Ratio:** If two numbers are in ratio, then the ratio of their cubes is called triplicate ratio. If x and y are two numbers, then the triplicate ratio of x and y would be $x^3 : y^3$.
For example: Triplicate ratio of $2:3 = 2^3 : 3^3 = 8:27$

4. **Sub-triplicate Ratio:** If two numbers are in ratio, then the ratio of their cube roots is called sub-triplicate ratio. If x and y are two numbers, then the sub-triplicate ratio of x and y would be $\sqrt[3]{x} : \sqrt[3]{y}$.
For example: Sub-triplicate ratio of $1:125 = \sqrt[3]{1} : \sqrt[3]{123} = 1:5$

5. **Inverse Ratio:** If two numbers are in ratio, then their antecedent and consequent are interchanged and the ratio obtained is called inverse ratio, If x and y are two numbers and their ratio is $x : y$, then its inverse ratio will be $y : x$.
For example: Inverse ratio of $4:5$ is $5:4$.

6. **Compound Ratio:** If two or more ratios are given, then the antecedent of one is multiplied with antecedent of other and respective consequents are also multiplied. If $a : b, c : d$ and $e : f$ are three ratios, then their compound ratio will be $ace : bdf$.
For example: The compound ratio of 2:5, 6:7 and 9:13 $= \dfrac{2 \times 6 \times 9}{5 \times 7 \times 13} = \dfrac{108}{455}$

Note

(i) If the antecedent is greater than the consequent, then the ratio is known as the ratio of greater inequality, such as $7:5$

(ii) If the antecedent is less than the consequent, then the ratio is called the ratio of less inequality, such as $5:7$

8.1.2 Comparison of Ratios

Rules used to compare different ratios are as follows

Rule I

If the given ratios are $a : b$ and $c : d$, then

(i) $a : b > c : d$, if $ad > bc$

(ii) $a : b < c : d$, if $ad < bc$

(iii) $a : b = c : d$, if $ad = bc$

Ex. 1 Which is greater $\dfrac{5}{8}$ or $\dfrac{9}{14}$?

Sol. Let $\dfrac{a}{b} = \dfrac{5}{8}$ and $\dfrac{c}{d} = \dfrac{9}{14}$

Then $ad = 5 \times 14 = 70$ and $bc = 8 \times 9 = 72$

$$\because ad > bc$$

$$\therefore \frac{a}{b} > \frac{c}{d} \implies \frac{5}{8} > \frac{9}{14}$$

Rule II

If two ratios are given for comparison, convert each ratio in such a way that both ratios have same denominator, then compare their numerators, the fraction with greater numerator will be greater. Ex. 2 Find the greater ratio between 2 : 3 and 4 : 5.

Sol. $\frac{2 \times 5}{3 \times 5} = \frac{10}{15}$ and $\frac{4 \times 3}{5 \times 3} = \frac{12}{15}$ $\left[\because \text{LCM of denominator of } \frac{2}{3} \text{ and } \frac{4}{5} \; i.e, \text{ 3 and 5 is 15}\right]$

$$\implies \frac{12}{15} > \frac{10}{15}$$

$$\therefore 4 : 5 > 2 : 3$$

Rule III

If two ratios are given for comparison, convert each ratio in such a way that both ratios have same numerator, then compare their denominators, the fraction with lesser denominator will be greater.

Ex. 3 Find the least fraction between $\frac{6}{7}$ and $\frac{7}{9}$

Sol. $\frac{6 \times 7}{7 \times 7} = \frac{42}{49}$ and $\frac{7 \times 6}{9 \times 6} = \frac{42}{56}$ $\left[\because \text{LCM of numerator of } \frac{6}{7} \text{ and } \frac{7}{9} \; i.e, \text{ 6 and 7 is 42}\right]$

$$\implies \frac{42}{49} > \frac{42}{56}$$

$$\therefore 6 : 7 > 7 : 9$$

8.2 Proportion

An equality of two ratios is called **Proportion**. If $\frac{a}{b} = \frac{c}{d}$ or $a : b = c : d$, the we say that a, b, c and d are in proportion and can be written as $a : b :: c : d$, where symbol$'$ ::$'$ represents proportion and it is read as 'a is to b' as 'c is to d.'

Here, a and d are called '**Extremes**' and b and c are called as '**Means**'.

8.2.1 Basic Rules of Proportion

Rule I

If $a : b :: b : c$, then c is called third proportional to a and b, which are in continued proportion, c will be calculated as

$$a : b :: b : c \implies a : b = b : c \implies a \times c = b \times b \implies ac = b^2 \implies c = \frac{b^2}{a}$$

Ex. 4 Calculate the 3rd proportional to 16 and 32.

Sol. Let 3rd proportional be x.

$$\text{Then,} \quad 16 : 32 :: 32 : x$$
$$\implies \quad \frac{16}{32} = \frac{32}{x}$$
$$\therefore \quad x = \frac{32 \times 32}{16} = 64$$

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Rule II

If $a : b :: c : d$, then d is called the 4th proportional to a, b and c. d will be calculated as

$$a : b :: c : d = \implies a : b = c : d \implies a \times d = c \times b \implies d = \frac{bc}{a}$$

Ex. 5 Find the 4th proportional to 3,7 and 9.

Sol. Let 4rd proportional be x.

$$\text{Then,} \quad 3 : 7 :: 9 : x \implies \frac{3}{7} = \frac{9}{x}$$
$$\implies \quad 3x = 9 \times 7 \implies x = \frac{9 \times 7}{3} = 21$$

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Rule III

Mean proportional between a and b is $\sqrt{ab}$. If mean proportional is x, then

$$a : x :: x : b$$
$$\implies \quad x^2 = a \times b$$
$$\implies \quad x = \sqrt{ab}$$

Ex. 6 What will be the mean proportional between 4 and 25?

Sol. Let mean proportional be x

$$\text{Then, } 4 : x :: x : 25$$
$$\implies 4 : x = x : 25$$
$$\implies 4 \times 25 = x \times x$$
$$\implies x = \sqrt{4 \times 25} = 10$$

Invertendo, Alternendo, Componendo and Dividendo

If $\frac{a}{b} = \frac{c}{d}$, then

1. Invertendo $\longrightarrow \frac{b}{a} = \frac{d}{c}$
2. Alternendo $\longrightarrow \frac{a}{c} = \frac{b}{d}$
3. Componendo $\longrightarrow \frac{a+b}{b} = \frac{c+d}{d}$
4. Dividendo $\longrightarrow \frac{a-b}{b} = \frac{c-d}{d}$
5. Componendo and Dividendo $\longrightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$
6. If $\frac{a}{b} = \frac{c}{d} = \frac{i}{j} = \ldots = k$ then, $\frac{a+c+i+\ldots}{b+d+j+\ldots} = k$

Ex. 7 Find the value of $\frac{a+b}{a-b}$, if $\frac{a}{b} = \frac{5}{3}$

Sol. Given, $\frac{a}{b} = \frac{5}{3}$

By Componendo and Dividendo,

$$\frac{a+b}{a-b} = \frac{5+3}{5-3} = \frac{8}{2} = 4$$

Ex. 8 If $\frac{3}{2} = \frac{18}{12} = \frac{24}{16} = \alpha$, then find the value of α.

Sol.

$$\alpha = \frac{\text{Sum of numerators}}{\text{Sum of denominators}} = \frac{3+18+24}{2+12+16} = \frac{45}{30} = \frac{3}{2}$$

Fast Track Techniques to solve the QUESTIONS

Technique 1

(i) If $A:B=a:b$ and $B:C=m:n$, then $A:B:C=am:mb:nb$ and $A:C=am:bn$

(ii) If $A:B=a:b$, $B:C=c:d$ and $C:D=e:f$, then
$A:B:C:D=ace:bce:bde:bdf$

Ex. 9 If $a:b=5:14$ and $b:c=7:3$, then find $a:b:c$.

Sol. $a:b:c=(5\times 7):(7\times 14):(14\times 3)$ [dividing the ratio by 7]
$=5:14:(2\times 3)=5:14:6$

Ex.10 The ratio of $A:B=1:3$, $B:C=2:5$ and $C:D=2:3$. Find the value of $A:B:C:D$.

Sol. $A:B=1:3, B:C=2:5, C:D=2:3$
$\therefore\ A:B:C:D=(1\times 2\times 2):(3\times 2\times 2):(3\times 5\times 2):(3\times 5\times 3)=4:12:30:45$

Technique 2

(i) If x is divided in $a:b$, then 1st part $=\dfrac{ax}{a+b}$; 2nd part $=\dfrac{bx}{a+b}$.

(ii) If x is divided in $a:b:c$, then 1st part $=\dfrac{ax}{a+b+c}$; 2nd part $=\dfrac{bx}{a+b+c}$; 3rd part $==\dfrac{cx}{a+b+c}$.

Ex. 11 Divide 1111 in the ratio of 8:3.

Sol. Let First part be $8x$; 2nd be $3x$

According to the equation,

$$8x+3x=1111 \implies 11x=1111$$

$$\therefore \quad x=\frac{1111}{11}=101$$

Now, First part $=8x=8\times 101=808$

Second Part $=3x=3\times 101=303$

Fast Track Method

$$\text{1st Part}=\frac{8}{8+3}\times 1111=\frac{8}{11}\times 1111=8\times 101=808$$

$$\text{2nd Part}=\frac{3}{8+3}\times 1111=\frac{3}{11}\times 1111=3\times 101=303$$

Ex. 12 Divide 2324 in the ratio of 35:28:20

Sol. $\text{1st Part} = \dfrac{35}{25+28+20} \times 2324 = \dfrac{35}{83} \times 2324 = 35 \times 28 = 980$

$\text{2nd Part} = \dfrac{28}{25+28+20} \times 2324 = \dfrac{28}{83} \times 2324 = 28 \times 28 = 784$

$\text{3rd Part} = \dfrac{20}{25+28+20} \times 2324 = \dfrac{20}{83} \times 2324 = 20 \times 28 = 560$

Ex. 13 The sum of three numbers is 315. If the ratio between 1st and 2nd is 2 : 3 and the ratio between 2nd and 3rd is 4 : 5, then find the 2nd number.

Sol. $\text{1st number : 2nd number} = 2:3 = (2\times4):(3\times4) = 8:12$

$\text{2nd number : 3rd number} = 4:5 = (4\times3):(5\times3) = 12:15$

$\therefore$ $\text{1st number : 2nd number : 3rd number} = 8:12:15$

$$\text{2nd number} = \frac{12}{8+12+15} \times 315 = \frac{12}{35} \times 315 = 108$$

Technique 3

The incomes of two persons are in ratio of $a:b$ and their expenditures are in the ratio of $c:d$. If each of them saves ₹X, then their incomes are given by

$$\frac{X(d-c)}{ad-bc} \times a \qquad \text{and} \qquad \frac{X(d-c)}{ad-bc} \times b$$

respectively and their expenditures are given by

$$\frac{X(b-a)}{ad-bc} \times c \qquad \text{and} \qquad \frac{X(b-a)}{ad-bc} \times d$$

respectively.

Ex.14 The ratio of incomes of Raman and Gagan is 4 : 3 and ratio of their expenditures is 3 : 2. If each person saves ₹2500, then find their incomes and expenditures.

Sol. Let the income of Raman be ₹$4x$ and that of Gagan be ₹$3x$.

Expenditure of Raman = ₹$(4x-2500)$, Expenditure of Gagan = ₹$(3x-2500)$

According to the question,

$$\frac{4x-2500}{3x-2500} = \frac{3}{2}$$

$$\Rightarrow 8x-5000 = 9x-7500 \Rightarrow x = 7500-5000 = 2500$$

Income of Raman = $4x = 4\times2500 =$ ₹ 10000

Income of Gagan = $3x = 3\times2500 =$ ₹ 7500

Expenditure of Raman = $4x-2500 = 10000-2500 =$ ₹ 7500

Expenditure of Gagan = $3x-2500 = 7500-2500 =$ ₹ 5000

Fast Track Method

Here, $a = 4,\ b = 3,\ c = 3,\ d = 2$, and $X = ₹\,2500$

Income of Raman:

$$\frac{X(d-c)}{ad-bc} \times a = \frac{2500(2-3)}{8-9} \times 4 = ₹\,10000$$

Income of Gagan:

$$\frac{X(d-c)}{ad-bc} \times b = \frac{2500(2-3)}{8-9} \times 3 = ₹\,7500$$

Expenditure of Raman:

$$\frac{X(b-a)}{ad-bc} \times c = \frac{2500(3-4)}{8-9} \times 3 = ₹\,7500$$

Expenditure of Gagan:

$$\frac{X(b-a)}{ad-bc} \times d = \frac{2500(3-4)}{8-9} \times 2 = ₹\,5000$$

Technique 4

If two numbers are in ratio $a : b$ and x is added to the numbers, then the ratio becomes $c : d$. The two numbers will be

$$\frac{xa(c-d)}{ad-bc} \text{ and } \frac{xb(c-d)}{ad-bc}$$

Ex. 15 Two numbers are in the ratio of 2:3. If 15 is added to both numbers, then the ratio between two numbers become $\frac{11}{14}$. Find the greater number.

Sol. Let the numbers be $2x$ and $3x$.

According to the formula,

$$\frac{2x+15}{3x+15} = \frac{11}{14}$$

$$\implies 14(2x+15) = 11(3x+15)$$

$$\implies 28x + 210 = 33x + 165$$

$$\implies 33x - 28x = 210 - 165$$

$$\implies 5x = 45 \implies x = \frac{45}{5} = 9$$

$\therefore$ The greater number $= 3 \times 9 = 27$

Fast Track Method

Here, $a = 2, b = 3, c = 11, d = 14$ and $x = 15$

Then, the two numbers are

$$\frac{xa(c-d)}{ad-bc} \quad \text{and} \quad \frac{xb(c-d)}{ad-bc}$$

$$\Longrightarrow \quad \frac{15\times 2(11-14)}{2\times 14-(3\times 11)} \quad \text{and} \quad \frac{15\times 3(11-14)}{2\times 14-(3\times 11)}$$

$$\Longrightarrow \quad \frac{30\times(-3)}{28-33} \quad \text{and} \quad \frac{45\times(-3)}{28-33}$$

$$\Longrightarrow \quad \frac{30\times(-3)}{-5} \quad \text{and} \quad \frac{45\times(-3)}{-5}$$

$$\Longrightarrow \quad 18 \quad \text{and} \quad 27$$

$\therefore$ The greater number is 27.

Technique 5

Two numbers are in the ratio $a:b$ and x is subtracted from the numbers, then the ratio is $c:d$. The two numbers will be

$$\frac{xa(d-c)}{ad-bc} \text{ and } \frac{xb(d-c)}{ad-bc}$$

Ex. 16 Two numbers are in the ratio of 3 : 5. If 9 is subtracted from each, the ratio becomes 12 : 23. Find the greater number.

Sol. Here, $a=3, b=5, c=12, d=23$ and $x=9$

$$\text{Then, 1st number} = \frac{xa(d-c)}{ad-bc} = \frac{9\times 3(23-12)}{3\times 23-(5\times 12)} = \frac{27\times 11}{69-60} = \frac{27\times 11}{9} = 3\times 11 = 33$$

$$\text{2nd number} = \frac{xa(d-c)}{ad-bc} = \frac{9\times 5(23-12)}{3\times 23-(5\times 12)} = \frac{45\times 11}{69-60} = \frac{45\times 11}{9} = 5\times 11 = 55$$